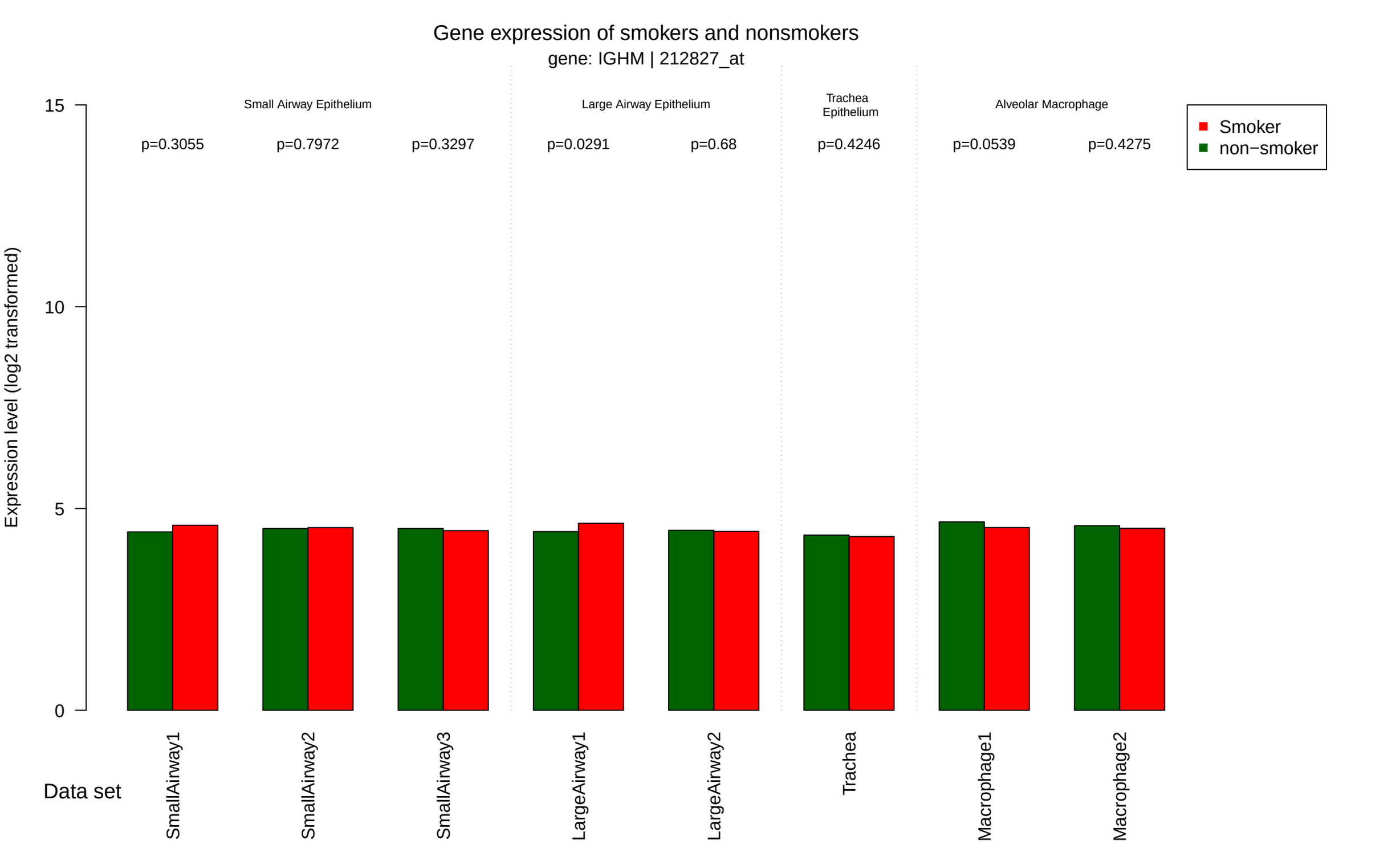
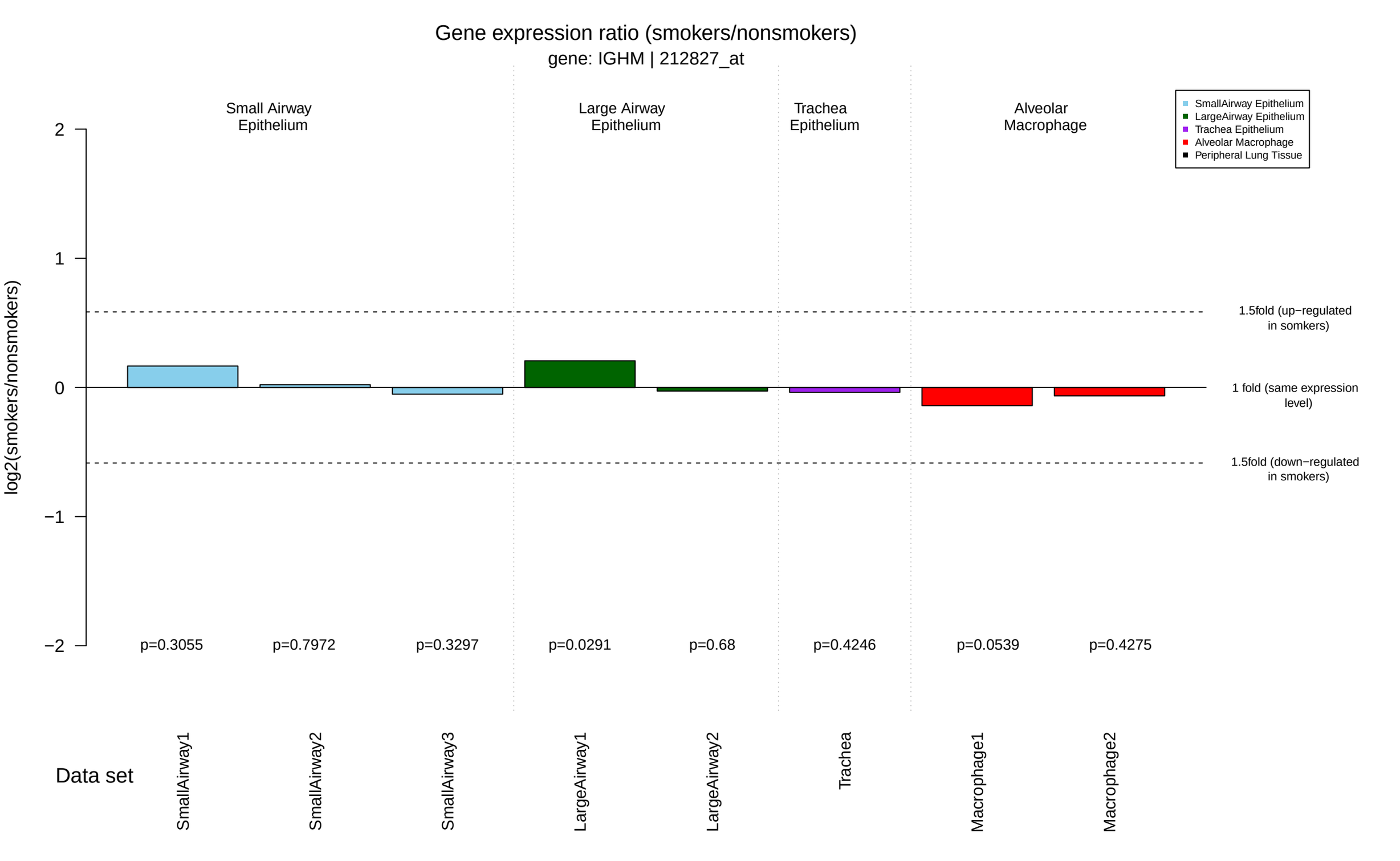
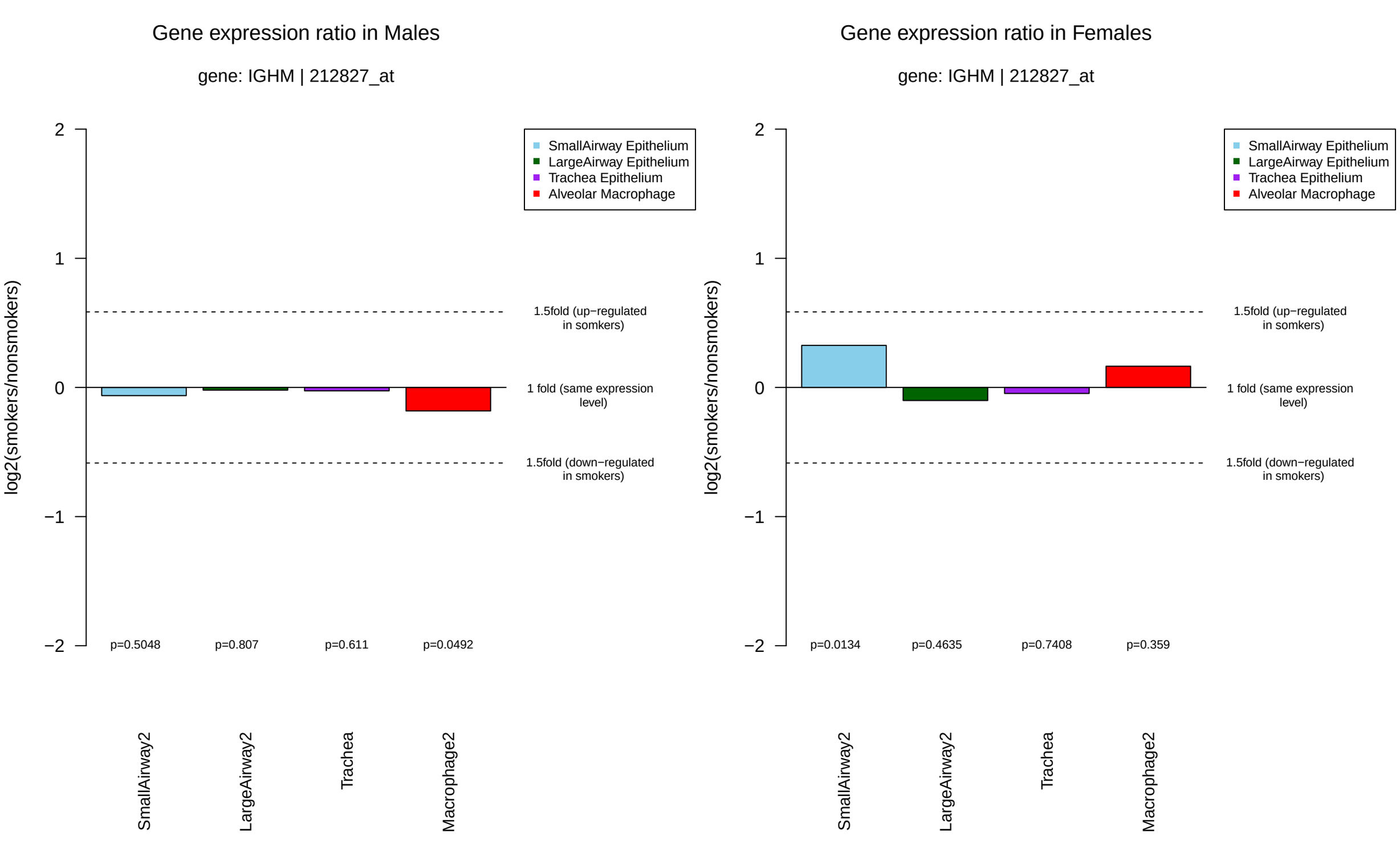
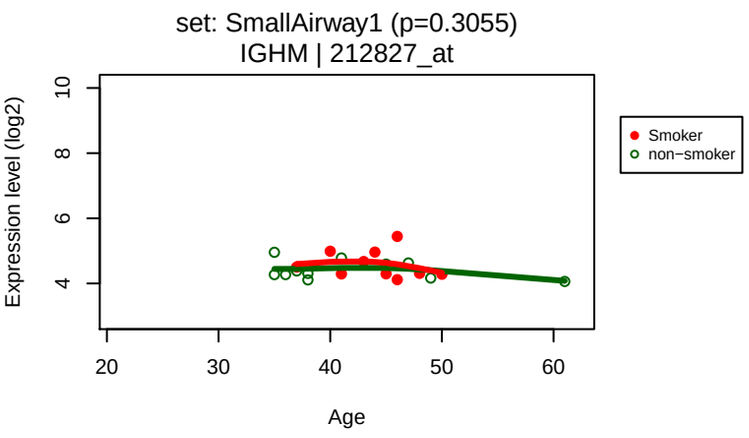
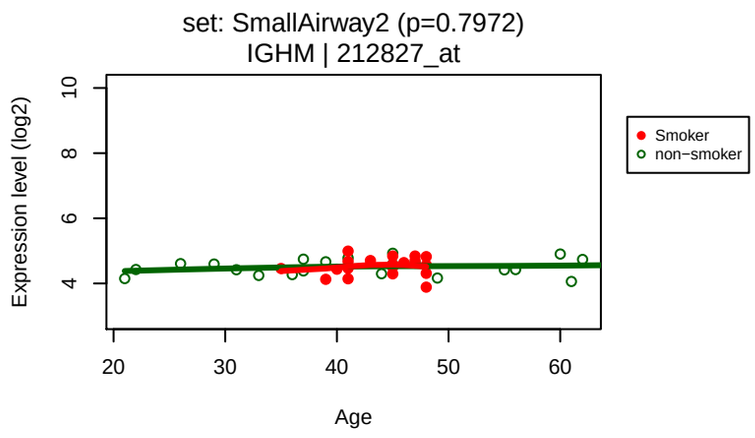
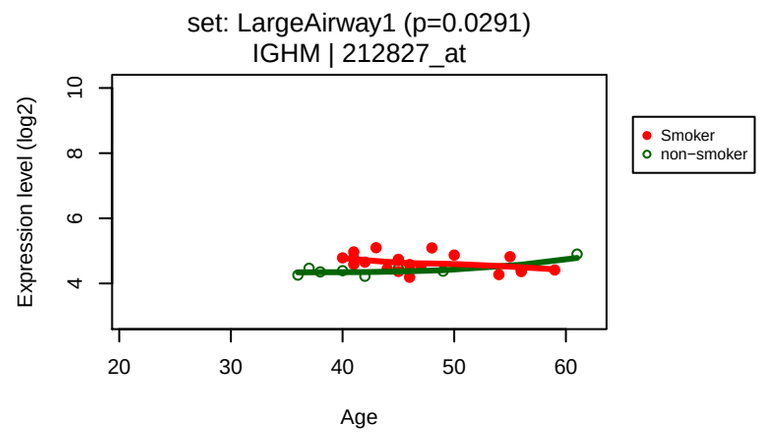

Gene expression of smokers and nonsmokers
gene: IGHM | 212827_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3055
p=0.7972
p=0.3297
p=0.0291
p=0.68
p=0.4246
p=0.0539
p=0.4275
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

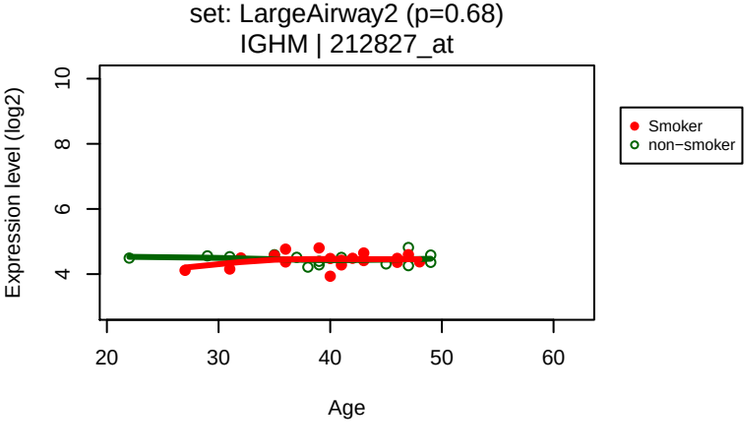

Gene expression ratio (smokers/nonsmokers)
gene: IGHM | 212827_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3055
p=0.7972
p=0.3297
p=0.0291
p=0.68
p=0.4246
p=0.0539
p=0.4275
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

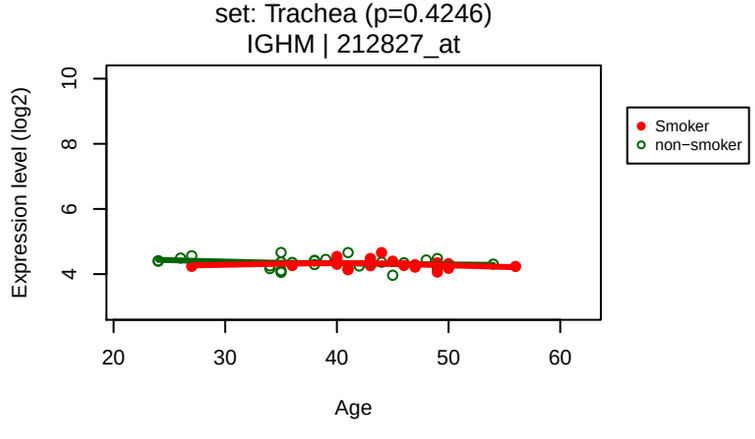

Gene expression ratio in Males
gene: IGHM | 212827_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5048
p=0.807
p=0.611
p=0.0492
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHM | 212827_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0134
p=0.4635
p=0.7408
p=0.359
SmallAirway2
LargeAirway2
Trachea
Macrophage2

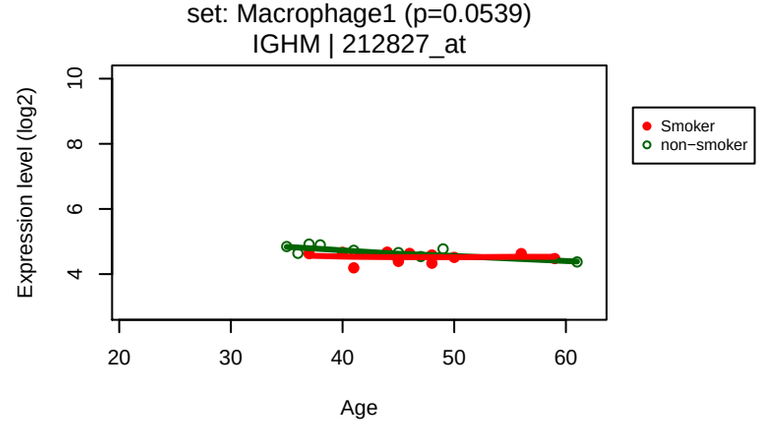

set: SmallAirway1 (p=0.3055)
IGHM | 212827_at
Expression level (log2)
Age
Smoker
non−smoker

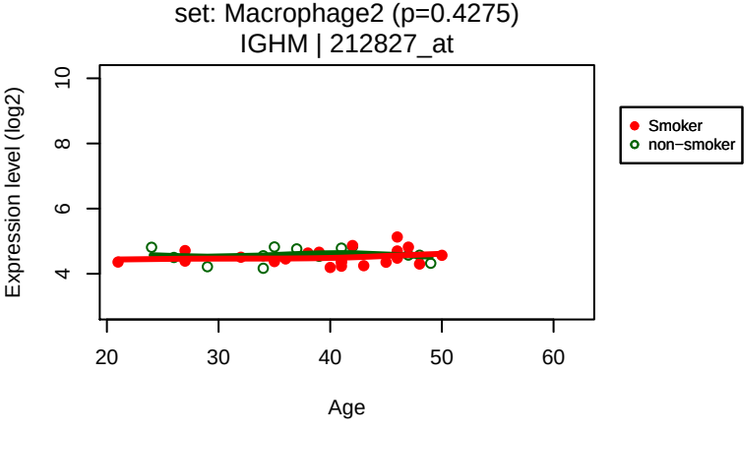

set: SmallAirway2 (p=0.7972)
IGHM | 212827_at
Expression level (log2)
Age
Smoker
non−smoker

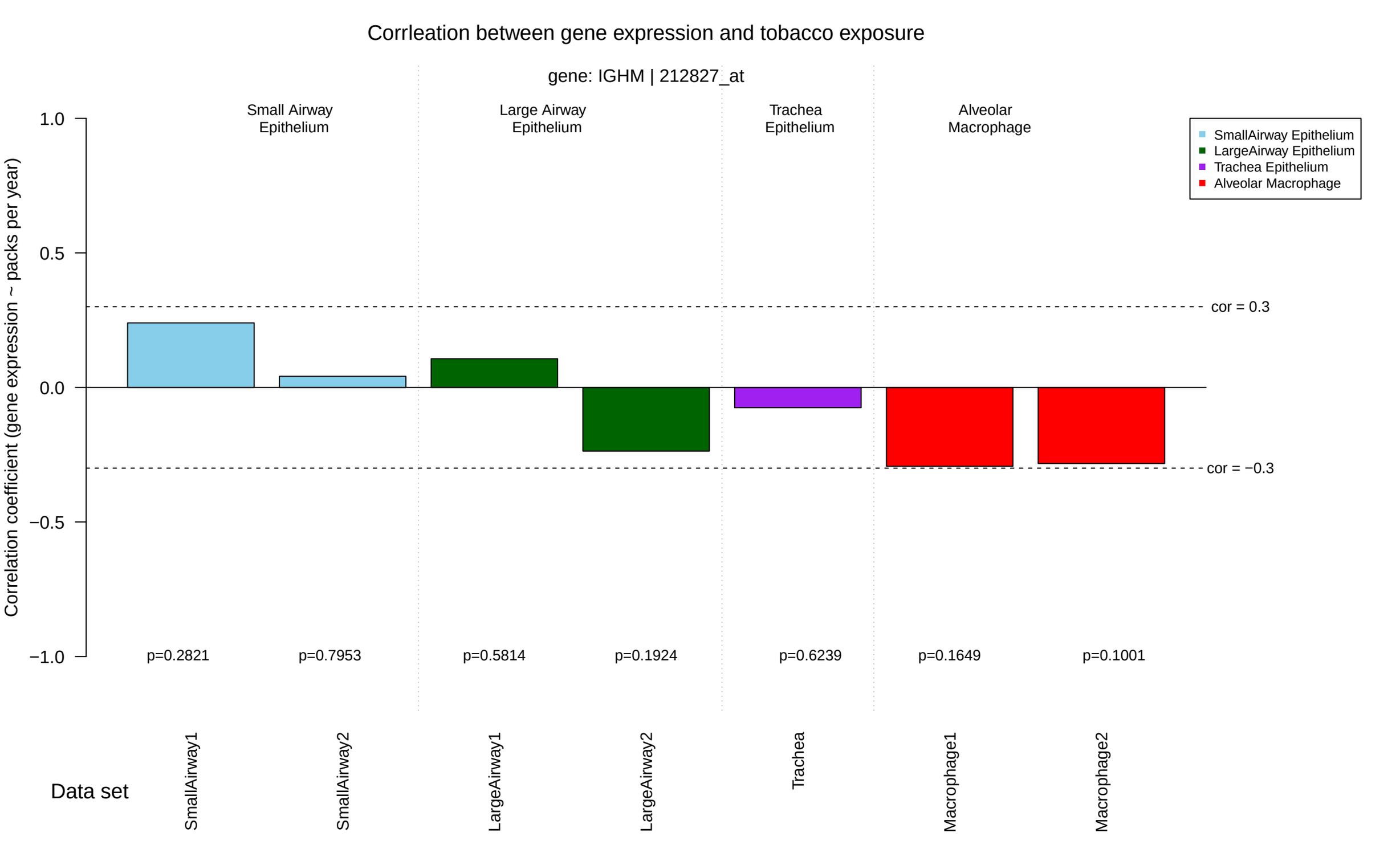

set: LargeAirway1 (p=0.0291)
IGHM | 212827_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.68)
IGHM | 212827_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4246)
IGHM | 212827_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0539)
IGHM | 212827_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4275)
IGHM | 212827_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHM | 212827_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2821
p=0.7953
p=0.5814
p=0.1924
p=0.6239
p=0.1649
p=0.1001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2